

Learning to Collaborate: An Orchestrated-Decentralized Framework for Peer-to-Peer LLM Federation

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Abstract

Fine-tuning Large Language Models (LLMs) for specialized domains is constrained by a fundamental challenge: the need for diverse, cross-organizational data conflicts with the principles of data privacy and sovereignty. While Federated Learning (FL) provides a framework for collaboration without raw data exchange, its classic centralized form introduces a single point of failure and remains vulnerable to model inversion attacks. Decentralized FL (DFL) mitigates this risk by removing the central aggregator but typically relies on inefficient, random peer-to-peer (P2P) pairings, forming a collaboration graph that is blind to agent heterogeneity and risks negative transfer. This paper introduces KNEXA-FL, a novel framework for *orchestrated decentralization* that resolves this trade-off. KNEXA-FL employs a non-aggregating Central Profiler/Matchmaker (CPM) that formulates P2P collaboration as a contextual bandit problem, using a LinUCB algorithm on abstract agent profiles to learn an optimal match-making policy. It orchestrates direct knowledge exchange between heterogeneous, PEFT-based LLM agents via secure distillation, without ever accessing the models themselves. Our comprehensive experiments on a challenging code generation task show that KNEXA-FL yields substantial gains, improving Pass@1 by $\approx 50\%$ relative to random P2P collaboration. Critically, our orchestrated approach demonstrates stable convergence, in stark contrast to a powerful centralized distillation baseline which suffers from catastrophic performance collapse. Our work establishes adaptive, learning-based orchestration as a foundational principle for building robust and effective decentralized AI ecosystems.

Code — <https://github.com/FujitsuResearch/knexa-fl>

Introduction

Adapting Large Language Models (LLMs) to specialized domains presents a fundamental dilemma: the need for diverse, cross-organizational data clashes with inviolable principles of data sovereignty, privacy, and security. While Federated Learning (FL) (McMahan et al. 2017) offers a paradigm for collaborative training without raw data exchange, its canonical server-centric form introduces a trusted central aggregator. This entity becomes a single point of

failure and a privileged attack surface for sophisticated model inversion attacks that can reconstruct sensitive training data (Geiping et al. 2020; Zhao et al. 2024).

Decentralized FL (DFL) addresses this by eschewing a central server in favor of direct peer-to-peer (P2P) communication (Roy et al. 2019; Itahara et al. 2023). However, in removing the central aggregator, DFL architectures typically regress to simplistic interaction strategies, such as random or static peer pairings. This random-walk approach to collaboration is statistically inefficient, blind to agent heterogeneity, and risks *negative transfer*, where poorly matched peers degrade each other’s performance. It fails to strategically harness the network’s latent collective intelligence.

This paper argues that the prevailing dichotomy between a vulnerable central aggregator and inefficient random P2P interaction is too limited. We introduce KNEXA-FL, a novel hybrid framework that enables *orchestrated decentralization*. It features a non-aggregating **Central Profiler/Matchmaker (CPM)** whose sole purpose is to learn an optimal matchmaking policy for a federation of heterogeneous, autonomous LLM agents. The CPM operates on abstract, privacy-preserving agent profiles to solve a contextual bandit problem, intelligently routing knowledge exchange without ever accessing or storing the models themselves. The actual knowledge transfer, via prediction distillation compatible with parameter-efficient fine-tuning (PEFT), occurs directly and securely between the matched peers.

Our contributions are threefold:

- We designed and propose KNEXA-FL, a hybrid, centrally orchestrated - decentralized learning architecture for collaborative LLM fine-tuning. It resolves the match-making inefficiency of DFL without re-introducing the security vulnerabilities of a central model aggregator, thus offering a more robust and effective collaboration paradigm.
- We are the first to formulate and solve the P2P LLM collaboration problem as a contextual bandit. Our CPM employs a LinUCB algorithm to learn a pairing policy from interaction rewards, dynamically optimizing the P2P knowledge graph to maximize network utility.
- We provide comprehensive empirical validation on a challenging, heterogeneous code generation task. Our experiments show that KNEXA-FL substantially outper-

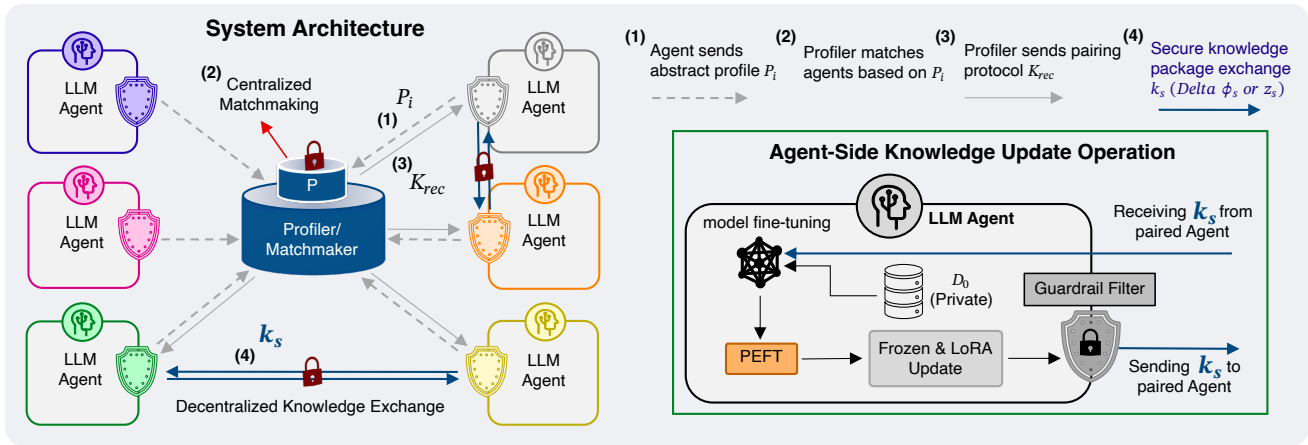


Figure 1: The KNEXA-FL architecture for orchestrated decentralization. Agents send abstract profiles ($\mathbf{p}_i$) to a non-aggregating Central Profiler/Matchmaker (CPM), which provides pairing directives (K_{rec}). Paired agents then conduct direct, secure P2P knowledge exchange (k_s). The agent-side view (right) details local PEFT fine-tuning and the privacy-enforcing Guardrail Filter.

forms isolated and random P2P baselines, and, critically, demonstrates stable convergence where a state-of-the-art centralized knowledge distillation approach suffers from catastrophic collapse.

Related Work

KNEXA-FL integrates ideas from four strands of literature: (i) decentralized / peer-to-peer federated learning, (ii) parameter-efficient LLM federation, (iii) security-aware governance in collaborative AI, and (iv) orchestration, data spaces, and multi-agent systems. We briefly position our contributions in each area.

Federated and Decentralized Learning

Federated Learning (FL) enables collaborative model training by exchanging parameter updates rather than raw data (McMahan et al. 2017). Canonical server-centric FL, however, struggles with statistical heterogeneity (non-IID data) and system heterogeneity (diverse model architectures). While methods like FedProx (Li et al. 2020) and SCAFFOLD (Karimireddy et al. 2020) address statistical drift, they maintain a central aggregator. Similarly, server-mediated knowledge distillation (e.g., FedMD (Li and Wang 2019)) handles architectural diversity but preserves the central server as a single point of failure and a potential privacy bottleneck.

Decentralized FL (DFL) obviates the server, typically relying on gossip-based protocols where agents average parameters with random neighbors (Hegedűs, Danner, and Jelasity 2021; Roy et al. 2019). Such methods generally assume model homogeneity and do not perform intelligent peer selection. More recent P2P frameworks have introduced greater flexibility through knowledge distillation (Itahara et al. 2023), sub-network exchange (Belal, Aksu, and Uluagac 2023), or topology-aware aggregation (Ryabinin et al. 2021). Despite these advances, the fundamental problem of *matchmaking* remains underdeveloped, often relying on random or handcrafted heuristics.

This is the critical gap our work addresses. In sharp contrast to prior art, our **CPM** formulates peer selection as a contextual bandit problem, learning to construct a high-utility interaction graph dynamically. Existing learning-based schedulers like Oort (Lai et al. 2021) and FedBalancer (Shin et al. 2022) optimize *server-client* selection for system metrics like throughput, a fundamentally different problem. The most relevant prior work, IPLS (Pappas et al. 2021), performs a *single, static peer grouping* at the outset based on model similarity. KNEXA-FL is the first framework to employ online learning to *continually* orchestrate the P2P collaboration graph, adapting to evolving agent knowledge while remaining fully non-aggregating.

Parameter-Efficient LLM Federation

The immense scale of LLMs has catalyzed research into federated fine-tuning using PEFT techniques. Centralized frameworks like FATE-LLM (Liu et al. 2023) and FedLoRA variants (Yang et al. 2024) federate PEFT modules (e.g., LoRA adapters) but rely on a central server for aggregation, which can lead to destructive parameter interference. Decentralized approaches such as FedSKD (Weng, Cai, and Zhou 2025) and KD-PDFL (Jeong and Kountouris 2023) use P2P knowledge distillation, but as noted, they lack an intelligent matchmaking mechanism. KNEXA-FL advances this line of work by uniquely combining support for heterogeneous LLM backbones and PEFT methods with CPM-guided P2P transfers, eliminating the risks of centralized aggregation while maximizing the efficacy of knowledge exchange.

Security and Governance in Collaborative AI

Security for FL and deployed ML systems covers AML risk analysis and robustness of perception/representation components (Bitton et al. 2023; Singh, Kakizaki, and Araki 2024), Byzantine-robust and cryptographic aggregation (Pillutla, Kakade, and Harchaoui 2022; Bonawitz et al. 2017), and decentralized trust via reputation or blockchain mechanisms (Kang et al. 2022; Katevas et al. 2020). However,

these approaches do not solve the problem of efficient partner selection. KNEXA-FL integrates governance directly into the learning process: the CPM’s bandit model naturally learns to down-weight low-utility or malicious peers by observing their associated rewards. This is complemented by agent-side *Guardrail Filters* that enforce local data sovereignty policies. This learning-based approach to governance, where trust is an emergent property of observed utility, distinguishes our work from systems with static, rule-based policies.

Orchestration, Data Spaces, and Multi-Agent Systems

The vision of International Data Spaces (Otto and et al. 2022) emphasizes data sovereignty through policy-governed exchange. KNEXA-FL provides a concrete mechanism to realize a *knowledge data space*, where the exchanged assets are not raw data but abstract knowledge representations (distilled predictions: logits or decoded text), governed by policies enforced at both the agent level (Guardrails) and the system level (CPM orchestration).

From a mathematical and systems perspective, our framework can be viewed through the lens of Multi-Agent Systems (MAS). The problem of finding effective collaborators is a classic challenge in MAS, often addressed via coalition formation (Georgio et al. 2025) or task allocation. However, much of this work does not contend with the unique constraints of federated learning, namely extreme statistical heterogeneity and strict privacy requirements. Incentive-driven FL (Zhan et al. 2021) uses economic mechanisms to encourage participation but does not typically solve the problem of *who* should collaborate with whom for maximal learning efficacy. KNEXA-FL recasts the MAS coordination problem for federated LLMs as a contextual bandit problem. This is a significant departure from prior art: instead of relying on predefined rules or complex negotiation protocols, we leverage online learning to solve the coordination problem directly, optimizing for empirical performance in a dynamic, privacy-critical environment.

The KNEXA-FL Framework

The KNEXA-FL framework facilitates the collaborative enhancement of heterogeneous Large Language Models (LLMs) by enabling direct, orchestrated knowledge exchange among autonomous agents. It operates as a decentralized knowledge data space, circumventing data or model centralization. Interactions are guided by a CPM that analyzes dynamic, abstract agent profiles to recommend optimal P2P pairings and knowledge exchange protocols. This combination of adaptive P2P mechanisms with a learning-based central orchestrator addresses the dual challenges of LLM heterogeneity and effective, privacy-preserving knowledge sharing.

Problem Setting and System Architecture

We consider a dynamic set of N_t LLM agents $\mathcal{A}_t = \{a_1, \dots, a_{N_t}\}$ at round t . Each agent a_i holds a frozen

base model W_0 and a trainable PEFT module ϕ_i , primarily LoRA (Hu et al. 2021), on its private non-IID dataset D_i . The collective goal is a constrained multi-objective optimization:

$$\min_{\{\phi_i\}} \sum_{i=1}^{N_t} w_i \mathbb{E}_{(x,y) \sim D_i} [\ell(M_i(x), y)] \quad (1)$$

subject to a set of constraints including: privacy leakage, communication, and computation budgets.

The system architecture, illustrated in Figure 1, comprises three logical components. **(1) LLM Agents ($\mathcal{A}$)** are autonomous entities, each operating via a Local Gateway that enforces local policies and includes a Guardrail Filter to prevent sensitive data disclosure. **(2) The CPM ($\mathcal{P}$)** is a trusted, non-aggregating entity that receives abstract profiles $\mathbf{p}_i$ to orchestrate P2P interactions without accessing raw data or models. **(3) Secure P2P Communication Channels** are established directly between paired agents for ephemeral, encrypted knowledge exchange. This architecture establishes a managed data space where agents are sovereign knowledge providers, and $\mathcal{P}$ is the intelligent interaction broker.

P2P Collaboration Protocol

The KNEXA-FL protocol unfolds in iterative rounds of local training, profiling, matchmaking, and P2P knowledge exchange.

Agent-Side Operations. Each agent a_i first performs local PEFT updates by fine-tuning its module ϕ_i on its private data D_i : $\phi_i \leftarrow \phi_i - \eta_L \nabla_{\phi_i} \mathcal{L}_i(W_0, \phi_i; D_i)$. Subsequently, it prepares a knowledge package κ_i for sharing. For Adaptive Knowledge Distillation (AKD), this package contains *teacher predictions* produced on a shared, privacy-vetted transfer set $\mathcal{X}_u$. These predictions may be represented as logits $\mathbf{z}_i(\mathbf{x})$ or as decoded text sequences $y_i(\mathbf{x})$; in our implementation we use the latter. All payloads are vetted by the agent’s Guardrail Filter before packaging.

Adaptive Knowledge Distillation (AKD). AKD, our core exchange mechanism, uses **text-based distillation** for robustness to architectural and tokenizer heterogeneity. A teacher agent a_j generates text predictions $y_j(x)$ for each prompt $x \in \mathcal{X}_u$, which are sent to student a_i . The student a_i re-encodes the teacher’s text $y_j(x)$ with its own tokenizer to create a “soft” target token sequence $\tilde{y}_j(x)$. It then minimizes a standard token-level cross-entropy loss, forcing its own output distribution $p_i(\cdot | x)$ to match the teacher’s sequence:

$$\mathcal{L}_{\text{total},i}^{\text{KD}} = (1 - \alpha_{\text{KD}}) \mathcal{L}_i(D_i) + \alpha_{\text{KD}} \mathbb{E}_{x \in \mathcal{X}_u} [\mathcal{L}_{\text{CE}}(\tilde{y}_j(x), p_i(\cdot | x))], \quad (2)$$

where $\mathcal{L}_{\text{CE}}$ is the token-level cross-entropy loss w.r.t. the re-encoded teacher tokens $\tilde{y}_j(x)$. This text-level approach makes the objective well-defined for any pair, sidestepping all tokenizer mismatch issues.

The Central Profiler/Matchmaker (CPM)

The CPM is the learning-based orchestrator. It intelligently pairs agents by solving a contextual combinatorial bandit problem, moving beyond random or heuristic matchmaking.

Method	Core Architecture			Key Capabilities		Focus Area	
	Decentral. Arch.	P2P Exchange	Adaptive Matchmaking	Heterog. Support	Governance	Theory	LLM-Native
KNEXA-FL (ours)	✓	✓	✓ (<i>Learned</i>)	✓	✓	✓	✓
<i>Centralized Schedulers</i>							
Oort (Lai et al. 2021)	—	—	✓ (<i>Learned</i>)	—	(✓)	—	—
<i>Decentralized Systems</i>							
GossipLearn (Hegedűs, Danner, and Jelasity 2019)	✓	✓	—	—	(✓)	✓	—
IPLS (Pappas et al. 2021)	(✓)	✓	(✓) (<i>Static</i>)	(✓)	—	—	—
SparSFA (Wang et al. 2023)	✓	✓	—	—	✓	—	—
KD-PDFL (Jeong and Kountouris 2023)	✓	✓	—	✓	—	—	—
FedSKD (Weng, Cai, and Zhou 2025)	✓	✓	—	✓	—	—	(✓)
<i>Centralized Baselines</i>							
FedMD (Li and Wang 2019)	—	(✓)	—	✓	—	—	(✓)

Table 1: Comparison with representative FL systems. KNEXA-FL is the first to integrate learned adaptive matchmaking into a decentralized, heterogeneous, LLM-native P2P framework. Legend: ✓ = fully; (✓) = partial; — = not addressed.

Mechanism	Comm. Cost	Comp. Overhead	Heterogeneity Tol.	Info Specificity / Privacy Risk
AKD (Teacher predictions)	Med-High(logits)/Low(text)	Med-High (student train)	High	Medium
PEFT Module ($\Delta\phi$)	Low (e.g., MBs)	Low (Merge)	Low-Med (Needs T_{ij})	High (Specific param. changes)

Table 2: Comparison of Knowledge Exchange Mechanisms. AKD is the primary mechanism in KNEXA-FL due to its high heterogeneity tolerance and more favorable privacy-utility trade-off.

Agent Profiles and Contextual Bandit. Each agent a_i sends a privacy-preserving profile vector $\mathbf{p}_i \in \mathbb{R}^{d_p}$ to the CPM. This profile concatenates **static features** (e.g., LLM family, PEFT config), **dynamic features** (e.g., task performance, perplexity, privacy-preserving embeddings of local data distributions), and **historical/trust features** (e.g., success rates of past P2P interactions, CPM-maintained trust score). For a potential pair (a_i, a_j) , the CPM forms a context vector $\mathbf{x}_{ij}^{(t)} = \varphi(\mathbf{p}_i^{(t)}, \mathbf{p}_j^{(t)}, S_{net}^{(t)})$, capturing their compatibility and the global network state.

LinUCB-based Matchmaking. We employ LinUCB (Li et al. 2010) to select a set of K_p disjoint pairs per round that maximize expected utility. The CPM models the expected reward of a pairing as $\hat{r}_{ij} = \hat{\theta}^\top \mathbf{x}_{ij}$ and selects pairs based on an upper confidence bound (UCB) score to balance exploitation and exploration. The reward signal $r_{ij}^{(t)}$ provided by the receiving agent a_i after an exchange with a_j is a scalar value reflecting the utility of the interaction:

$$r_{ij}^{(t)} = \gamma (\mathcal{L}_i^{\text{pre}} - \mathcal{L}_i^{\text{post}}) - \delta \text{KB}_{ij}^{(t)}, \quad (3)$$

where the first term is the local loss reduction and the second penalizes communication cost ($\text{KB}_{ij}^{(t)}$). The CPM updates its bandit parameters ($\mathbf{A}, \mathbf{b}$) based on observed rewards, progressively learning the optimal matchmaking policy. The detailed matchmaking logic is presented in the Appendix.

Overall Protocol and Complexity

The complete KNEXA-FL protocol is specified in Algorithm 1. It formalizes the asynchronous, multi-phase loop involving parallel agent computation, centralized learning-based matchmaking, decentralized knowledge exchange, and the feedback mechanism that drives adaptation.

Complexity. The communication cost of a P2P exchange is dominated by the AKD payload. When sharing logits, this is $\mathcal{O}(|\mathcal{X}_u| \cdot |V|)$ bytes (e.g., FP16), which is about 62.5 MB for typical values. When sharing decoded text, the cost is $\mathcal{O}(|\mathcal{X}_u| \cdot L_{\text{avg}})$ tokens, which is substantially smaller in practice. The CPM’s computational overhead is modest. Each LinUCB feedback update is $\mathcal{O}(d_p^2)$, where d_p is the profile dimensionality. The matchmaking step, if naively enumerating all $\binom{N_t}{2}$ pairs, would be $\mathcal{O}(N_t^2 d_p)$. However, by pre-filtering candidates using efficient approximate nearest neighbor search on profile embeddings, the practical complexity is reduced to a tractable $\mathcal{O}(N_t k d_p)$ for a small neighborhood size $k \ll N_t$. Our implementation with 20 agents completes a full round in under 16 minutes on a cluster of eight A100 GPUs.

Security, Privacy, and Theoretical Insight

KNEXA-FL’s security is enhanced by several design principles. **Data Minimization** is achieved by exchanging only teacher predictions (logits or decoded text), never raw data. **Secure Communication** (e.g., mTLS with E2E payload encryption) ensures the CPM cannot decrypt knowledge pack-

Algorithm 1: The KNEXA-FL Protocol

```

1: Initialize: Profiler  $\mathcal{P}$  with LinUCB state ( $\mathbf{A} \leftarrow \mathbf{I}_{d_p}, \mathbf{b} \leftarrow \mathbf{0}$ ).
2: procedure AGENTUPDATE( $a_i, D_i, \phi_i$ )
3:    $\phi_i \leftarrow \phi_i - \eta_L \nabla_{\phi_i} \mathcal{L}_i(W_0, \phi_i; D_i) \triangleright$  Local PEFT update
4:    $\mathbf{p}_i \leftarrow \text{GenProfile}(\phi_i, D_i) \triangleright$  Generate abstract profile
5:   return  $\mathbf{p}_i$ 

6: for each communication round  $t = 1, 2, \dots, T$  do
  // Phase 1: Asynchronous Profiling
7:   for all agent  $a_i \in \mathcal{A}$  in parallel do
8:      $\mathbf{p}_i^{(t)} \leftarrow \text{AgentUpdate}(a_i, D_i, \phi_i^{(t-1)})$ 
9:     Send profile  $\mathbf{p}_i^{(t)}$  to  $\mathcal{P}$ .
  // Phase 2: Centralized Matchmaking
10:   $\mathcal{P}$  computes  $\hat{\boldsymbol{\theta}} \leftarrow \mathbf{A}^{-1}\mathbf{b}$  from its current state.
11:   $\mathcal{P}$  forms pairs  $\mathcal{E}_t = \{(a_s, a_r, \mathbf{x}_{sr})\}$  by selecting  $K_p$  disjoint pairs that greedily maximize the LinUCB score:  $\hat{\boldsymbol{\theta}}^\top \mathbf{x} + \beta \sqrt{\mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x}}$ .
12:   $\mathcal{P}$  dispatches matchmaking directives to agents in  $\mathcal{E}_t$ .
  // Phase 3 & 4: P2P Exchange and Policy Update
13:  for all pair  $(a_s, a_r, \mathbf{x}_{sr}) \in \mathcal{E}_t$  in parallel do
14:     $a_r$  receives knowledge package  $\kappa_s$  from  $a_s$  and integrates it via AKD (Eq. 2).
15:     $a_r$  computes reward  $r_{sr}^{(t)}$  via Eq. 3.
16:     $a_r$  reports feedback  $(\mathbf{x}_{sr}, r_{sr}^{(t)})$  to  $\mathcal{P}$ .
17:    On feedback receipt,  $\mathcal{P}$  updates its model:
18:     $\mathbf{A} \leftarrow \mathbf{A} + \mathbf{x}_{sr}(\mathbf{x}_{sr})^\top; \mathbf{b} \leftarrow \mathbf{b} + r_{sr}^{(t)} \mathbf{x}_{sr}$ .

```

ages. The **Non-Aggregating CPM** design mitigates central-point-of-failure risks. **Controlled Influence** is managed via the bandit, which learns to deprioritize malicious or low-quality peers, and through **Local Gateway Guardrail Filters** that scan outgoing knowledge packages for sensitive information. Future work can enhance **Verifiability** with DP-noise on logits or ZKPs for profile attestations.

Theoretically, the framework’s convergence can be analyzed by viewing the CPM as inducing a dynamic collaboration graph $\mathcal{G}_t$. The LinUCB regret bounds (Li et al. 2010) ensure the CPM efficiently learns to form graphs with high-utility edges (i.e., positive expected rewards). Drawing from decentralized consensus literature (Boyd et al. 2006), the system’s performance improvement is linked to the spectral properties (e.g., Fiedler value λ_2) of $\mathcal{G}_t$. As the CPM learns and adds more beneficial edges, $\mathbb{E}[\lambda_2(\mathcal{G}_t)]$ is non-decreasing, suggesting a trajectory towards monotonic performance improvement across the federation. We stress that this is an intuitive linkage to provide insight, not a formal convergence proof for the combined system.

Experiments

We conduct a comprehensive empirical study to validate KNEXA-FL, designed to answer three primary research

questions:

RQ1: Overall Performance Does profiler-guided P2P collaboration in KNEXA-FL achieve superior task performance compared to isolated training, unguided P2P collaboration, and a conventional centralized knowledge distillation baseline?

RQ2: Matchmaking Efficacy To what extent are the performance gains attributable to the intelligent matchmaking of the Central Profiler/Matchmaker over heuristic or random pairing strategies?

RQ3: Robustness & Scalability How does KNEXA-FL’s performance scale with the size of the federation and the degree of model and data heterogeneity among clients?

Experimental Setup

Datasets and Tasks. We focus on code generation, a challenging domain that demands complex reasoning and syntactic precision. We construct our primary dataset by merging the **HumanEval** (Chen et al. 2021) and **MBPP** (Austin et al. 2021) benchmarks, resulting in 464 unique programming problems. We create a 348/116 train/test split. To simulate realistic statistical heterogeneity, the 348 training problems are distributed among clients using a Dirichlet distribution with a concentration parameter $\alpha = 0.1$, ensuring a non-IID data landscape. An independent set of 128 problems is held out as the *knowledge-transfer set* ($\mathcal{X}_u$), used exclusively for knowledge distillation and unseen during local training.

Evaluation Metrics. Our primary metric is **Pass@k** ($k \in \{1, 5, 10\}$) (Chen et al. 2021), which measures the functional correctness of the generated code against unit tests. To assess syntactic and structural quality, we also report **CodeBLEU** (Ren et al. 2020).

Client Fleet and Models. To probe heterogeneity (RQ3), our main experiments feature a federation of **6** clients. Each client hosts a distinct open-source LLM backbone from a pool of models in the $\approx 500\text{M}$ parameter class, including Qwen1.5-0.5B, Cerebras-GPT-590M, bloom-560m, and pythia-410m. This 6-client setup is a deliberate stress test for high model/data heterogeneity, where naive collaboration fails and orchestration is most critical. All models are fine-tuned using LoRA (Hu et al. 2021), with ranks empirically optimized to keep trainable parameters between 2.2–3.0% of the total, significantly reducing communication payloads (details in Table 3).

Client	Backbone Model	# Params	Train %	Train/Val
C0	Qwen1.5-0.5B	475M	2.39%	45 / 12
C1	Cerebras-GPT-590M	604M	2.34%	43 / 11
C2	bloom-560m	572M	2.20%	44 / 12
C3	pythia-410m	418M	3.01%	46 / 12
C4	Qwen1.5-0.5B	475M	2.39%	54 / 14
C5	Cerebras-GPT-590M	604M	2.34%	44 / 11

Table 3: Heterogeneous 6-client configuration for our primary KNEXA-FL experiments. All backbones are tuned with empirically-optimized LoRA settings.

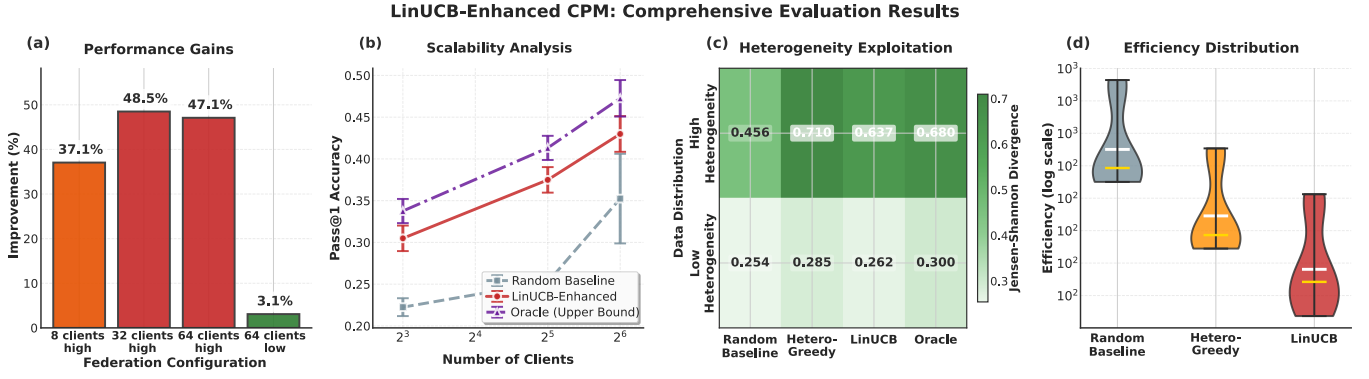


Figure 2: LinUCB-enhanced CPM comprehensive evaluation in a synthetic federation. (a) Performance Gains: The relative improvement over random pairing peaks at 48.5% for 32 clients in a high-heterogeneity setting and remains substantial (37.1%) even for 8 clients. Gains are modest (3.1%) in low-heterogeneity scenarios, confirming the CPM’s primary value is in exploiting diversity. (b) Scalability Analysis: Pass@1 accuracy demonstrates our LinUCB approach consistently outperforms the random baseline across all federation sizes (8 to 64 clients) and robustly tracks towards the oracle upper bound. (c) Heterogeneity Exploitation: The Jensen-Shannon (JS) divergence of selected pairs reveals the CPM’s learned strategy. While a naive Hetero-Greedy baseline maximizes JS divergence (0.710), our approach learns a superior trade-off, maintaining high divergence (≈ 0.64) while selecting for synergistic compatibility. (d) Efficiency Distribution: The computational efficiency, measured in rewards processed per second, remains highly practical, confirming that the CPM’s intelligence does not introduce prohibitive overhead. Error bars and shaded regions represent 95% confidence intervals over five independent runs.

Baselines. We compare KNEXA-FL against a rigorous set of baselines:

- **LocalOnly:** Establishes the performance frontier without collaboration. We report the average client performance after 12 rounds of isolated local fine-tuning.
- **FedID-CentralKD:** A modern centralized baseline, adapted from the Federated Interactive Distillation (FedID) framework (Ma et al. 2023). A server model (Qwen-0.5B) learns from client predictions on $\mathcal{X}_u$ via a confidence-weighted ensemble, then acts as a central teacher for all clients via text-level KD. This is privacy-preserving (server never sees private data).
- **Central-KD:** A strong centralized baseline inspired by FedMD (Li and Wang 2019). In each round, all clients submit their logits on the transfer set $\mathcal{X}_u$ to a central server. The server averages these logits to create an “ensemble teacher” distribution, which is then broadcast back to all clients for distillation. This represents a conventional approach to heterogeneous federated learning.
- **Heuristic-P2P:** A non-learning ablation using identical AKD. It replaces the CPM with a static heuristic: greedily pair clients to maximize data (JS) divergence, with the higher-performer as teacher.
- **Random-P2P:** A direct ablation of the CPM. This method uses the same P2P knowledge distillation as KNEXA-FL, but teacher-student pairings are selected uniformly at random in each of the 20 collaboration rounds.

Implementation Details. All experiments were conducted on a mix of NVIDIA H100s and A100s with clients simulated in parallel. Fixed random seed of 42 for reproducibility. Additional hyperparameters in the Appendix.

Results and Analysis

RQ1: Overall Performance. Table 4 validates our claims. The `LocalOnly` baseline (2.22% Pass@1) confirms collaboration is necessary. Our centralized baselines show extreme fragility under heterogeneity: the modern `FedID-CentralKD` failed to converge (1.11% Pass@1), reinforcing the `Central-KD` baseline’s collapse (peak 18.33% $\rightarrow$ final 2.00%) and confirming aggregation is unreliable in this regime. The P2P baselines are most critical. While `Random-P2P` (8.89% Pass@1) confirms P2P’s value, the `Heuristic-P2P` baseline, which maximizes data diversity (JS divergence), performed **worse** (6.67% Pass@1). This strongly suggests that naive, non-learning heuristics can be detrimental. In sharp contrast, **KNEXA-FL** achieves the highest performance (13.33% Pass@1), a **50%** relative gain over `Random-P2P` and **100%** over the failing heuristic. This demonstrates that our CPM’s *learned* policy, which balances diversity and compatibility (Fig. 2c), successfully navigates the trade-offs for stable, superior performance.

The Instability of Centralized Distillation. A deeper look at `Central-KD` shows fundamental instability. Although it briefly peaked at 18.33% Pass@1 (6 clients), a controlled 4-client run with full logging confirmed a collapse to 2.00% final Pass@1. Forcing heterogeneous models to distill from a single averaged “ensemble teacher” overwrites specialized knowledge, triggering catastrophic forgetting, which KNEXA-FL’s targeted, utility-driven P2P exchanges avoid.

RQ2: Matchmaking Efficacy. The performance gap between `Random-P2P` and KNEXA-FL directly points to the efficacy of the CPM. To isolate this effect, we mea-

Method	Pass@1 (%)	Pass@5 (%)	Pass@10 (%)	CodeBLEU
LocalOnly	2.22	5.42	5.55	0.260
FedID-CentralKD	1.11	5.56	5.56	0.181
Central-KD	2.00 (18.33) [†]	7.80	10.00	0.268
Heuristic-P2P [‡]	6.67	16.67	27.78	0.392
Random-P2P	8.89	22.40	27.80	0.239
KNEXA-FL	13.33	31.25	44.44	0.344

[†]Central-KD was volatile; peaked at 18.33% (6-client) but collapsed to 2.00% (4-client instability analysis). [‡]Heuristic-P2P was evaluated for representative restricted rounds and data on the same 6-client setup.

Table 4: Average performance on the global test set. LocalOnly is evaluated after 12 rounds of isolated training. Collaborative methods are evaluated after 20 rounds, unless otherwise noted. Best final performance is in bold.

sured the peak performance achieved by any student on the knowledge-transfer set during collaboration (Table 5). KNEXA-FL’s CPM-guided pairings enabled a student to achieve **86.70%** Pass@1 on this set, a staggering **2.6×** improvement over the best pairing found by Random-P2P. This demonstrates that the CPM is not merely avoiding bad pairings but is actively discovering and exploiting highly synergistic knowledge transfers that random chance is unlikely to find.

Pairing Strategy	Peak Student Pass@1
Random-P2P	33.33%
KNEXA-FL (CPM-Guided)	86.70%

Table 5: Peak student Pass@1 achieved on the 128-sample transfer set during collaboration. This metric isolates the quality of knowledge transfer.

Profiler Ablation: LinUCB under a Controlled Synthetic Regime The gains in Table 5 imply that *who* collaborates with whom is decisive. To attribute this effect precisely to the LinUCB-driven Central Profiler/Matchmaker and to test its scalability beyond the six-client, real-model setup, we conducted a controlled ablation in a large, **synthetic** environment, an established protocol for bandit and FL research (Lattimore and Szepesvári 2020).

The results, presented in Figure 2, are definitive:

- **Substantial and Scalable Gains:** The LinUCB-enhanced CPM delivers major performance improvements over random pairing, peaking at a **48.5%** relative gain in the 32-client, high-heterogeneity case (Panel a). This advantage scales robustly, with performance consistently approaching the oracle upper bound as the federation grows (Panel b).
- **Learned Compatibility Trade-off:** The CPM learns a non-trivial strategy for exploiting diversity. While a naive ‘Hetero-Greedy’ baseline maximizes Jensen-Shannon divergence, our CPM intelligently trades a small amount of diversity for a large gain in synergistic compatibility, explaining its superior performance (Panel c). This is

achieved with highly practical computational efficiency (Panel d).

Take-away. Under this isolated, reproducible setting, the LinUCB-driven CPM demonstrably *learns* a superior, scalable matchmaking policy that materially explains the end-to-end performance gains of KNEXA-FL, fully answering **RQ2**.

RQ3: Heterogeneous Federation. Per-client analysis of the 6-agent KNEXA-FL run shows that gains are distributed across the diverse federation. For instance, Client C2 (bloom-560m) evolved from a mid-tier performer to the strongest individual model, achieving a final local Pass@1 of 36.67%. Even the smallest model, C3 (pythia-410m), significantly surpassed its isolated performance, demonstrating that profiler-guided collaboration effectively lifts the entire ecosystem, not just the strongest members.

Discussion and Limitations

Our results show KNEXA-FL delivers substantial, stable gains with low overhead. The improvements stem from the CPM’s learned P2P orchestration, contrasting sharply with the volatile, “one-size-fits-all” centralized baselines. By targeting knowledge sharing, KNEXA-FL mitigates the catastrophic forgetting seen in simpler schemes.

Limitations. Our study has limitations. Future work should: (i) validate on larger federations with realistic WAN latencies; (ii) explore more semantic (e.g., user-profile-based) data splits beyond Dirichlet partitioning; and (iii) benchmark against a wider array of advanced centralized FL optimizers.

Despite these constraints, the consistent and stable performance edge over strong baselines firmly positions profiler-guided P2P learning as a robust and promising paradigm for decentralized, collaborative LLM intelligence.

Conclusion

We introduced KNEXA-FL, a framework for orchestrated decentralization that resolves the trade-off between insecure centralized FL and inefficient random P2P collaboration. Its non-aggregating Central Profiler/Matchmaker (CPM) formulates P2P matchmaking as a contextual bandit problem, learning to optimize the collaborative graph. Empirically, on a heterogeneous code-generation task, our approach yields substantial gains ($\approx 50\%$ relative Pass@1 improvement over random P2P) and, critically, achieves stable convergence where a strong centralized distillation baseline catastrophically fails. Our work establishes learning-based orchestration as a core principle for robust decentralized AI. Future work includes: (i) scaling to larger federations, (ii) exploring more expressive (e.g., neural) bandit models, and (iii) integrating safeguards such as differential privacy, zero-knowledge proofs, and token-efficient disparity audits like TFDp (Singh et al. 2025).

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